Mobile Starlink Performance over Recurring Route

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1 Problem

Low Earth Orbit (LEO) satellite networks like Starlink offer broadband connectivity almost anywhere, but their real world performance changes quite a bit once the terminal starts moving. Mobile terminals are now found mounted on buses, service vehicles, boats, and RVs, where users expect stable internet for navigation, telemetry, or remote work. However, vehicular deployments frequently suffer from sudden latency spikes, poor throughput, and brief connection dropouts.

The underlying causes of these mobile performance drops are not yet fully understood or explored. When a vehicle is in motion, network reliability depends on a mix of overhead obstacles (such as tree canopies, terrain cuts, and overpasses) and rapid orbital dynamics, where terminals must switch beams or satellites every few seconds. Because existing studies look at isolated drives, it is difficult to determine whether a bad connection is an unavoidable spot on the road, an inefficient satellite handover, or bad weather. Understanding where and why these dropouts occur is essential for setting realistic Quality of Service (QoS) expectations, planning routes with reliable connection, and designing smarter handover protocols for mobile satellite systems.

2 Prior Work

Laniewski et al. [2] conducted a comprehensive evaluation of mobile Starlink performance using a Flat High Performance (FHP) dish, mounted on a service van in Germany over a two-month period. They capturing over 15,000 stationary and mobile samples and found a mobile throughput degradation of 10%. Unfortunatly their dataset remains limited by a relatively short observation window.

Laniewski et al. [1] conducted a 300 km Autobahn test drive at fixed speeds up to 140 km/h. They measured signal degradation as the vehicle passed beneath 98 highway overpasses. Their study demonstrated significant performance degradation under high speed. Mobile packet loss increased by 814% and median RTT increased by 45% relative to stationary controls. Their tests highlights short term high speed with obstructions. The study is constrained by its brief timeframe.

Zhao et al. [3] investigated Starlink handovers under vehicular mobility. By analyzing telemetry from a 5-hour drive, they found that mobile terminals frequently break the standard 15-second handover interval to circumvent transient physical obstructions. These dynamic beam switching events cause severe network interruptions taking up to two seconds to recover. However, their short observation window cannot reveal long-term geographic or seasonal performance trends.

3 Approach

Most mobile Starlink studies rely on a single trip or a few weeks of data. In contrast, our approach takes advantage of a massive 1.5-year dataset collected from a school bus running the exact same two-hour route, twice a day, every weekday.

Because the bus drives the same road hundreds of times across different seasons, weather, traffic conditions, and use conditions, it acts as a natural controlled experiment. Holding the route constant makes it much easier to

figure out what actually causes bad performance, and can indicate whether a drop in service is tied to a specific spot on the road (like trees, hills, or bridges) or to satellite issues overhead.

Our approach focuses on three main steps:

- **Mapping Route Hotspots:** We map latency and packet loss along the entire route to see if specific spots consistently suffer from poor connectivity, and whether those trouble spots stay the same from day to day.
- **Comparing Trips Over Time:** We compare morning versus afternoon runs, weekdays versus weekends, and changes across seasons to see how time of day, network traffic, and weather affect performance.
- **Tracking Satellite Handovers:** Using logged satellite IDs and GPS, we track every time the terminal switches satellites to see how often handovers trigger latency spikes or temporary dropouts, and whether those handovers are scheduled or forced by blockages.

4 Deliverables

Below can be found our planned project deliverables and a rough biweekly schedule. Both the deliverables and schedule are subject to change.

Deliverables:

- Cleaned/processed dataset (latency + GPS + satellite ID aligned)

- Route-based visualizations / performance maps

- Handover analysis + latency-correlation results

- Statistical/ML model identifying recurring and anomalous patterns

- Final report + presentation

Biweekly Schedule:

- Weeks 1–2: data cleaning and exploration

- Weeks 3–4: Route-based visualizations / performance maps

- Weeks 5–6: Handover analysis + latency-correlation results

- Weeks 7–8: Statistical/ML model identifying recurring and anomalous patterns, Final Report

References

[1] Dominic Laniewski, Eric Lanfer, and Nils Aschenbruck. Measuring mobile starlink performance: A comprehensive look. *IEEE Open Journal of the Communications Society*, 6:1266–1283, 2025.

[2] Dominic Laniewski, Eric Lanfer, Simon Beginn, Jan Dunker, Michael Dückers, and Nils Aschenbruck. Starlink on the road: A first look at mobile starlink performance in central europe. In *2024 8th Network Traffic Measurement and Analysis Conference (TMA)*, pages 1–8, 2024.

[3] Jinwei Zhao, Jack Baude, Ali Ahangarpour, Vaibhava Krishna Devulapalli, Sree Ganesh Lalitaditya Divakarla, Zhi-Li Zhang, and Jianping Pan. Demystifying starlink network performance under vehicular mobility with dynamic beam switching. *arXiv preprint arXiv:2601.13790*, 2026.